Time to Build and the Macroeconomy

Robustness: union wage setting

Samuel Zhao
University of Chicago

October 24, 2026

The main text fixes the nominal wage at its steady-state level for the whole horizon. This note replaces the fixed wage with Rotemberg union wage setting, a standard sticky-wage block in New Keynesian models, and re-solves every policy experiment. Everything else is unchanged: the monthly calibration, fixed production lags, the shocks, the planner's problem (commitment, the zero lower bound, eight free months), and the comparison with the economy without time to build.

Summary.

- With 12-month wage contracts, about the average duration in U.S. data, the results are almost identical to the fixed wage. The monthly wage barely moves (by less than 0.01%).
- With much more flexible 3-month contracts, every qualitative result survives:
  - after a productivity fall, the planner cuts to zero with time to build and hikes to 25% annualised without it;
  - the zero bound binds after supply shocks with time to build, and after demand shocks without it;
  - policy matters far more for supply shocks with time to build.

  The ratio of gains is smaller, because flexible wages raise the gain without time to build.
- One result is specific to the fixed wage. With forward-looking unions, warning about a *demand* shock has some value even without time to build, because news reaches the economy through wages. For supply shocks, warning still has value only with time to build.

1 Union wage setting

Households supply differentiated labour to unions, which set nominal wages. The elasticity of substitution between labour types is ε_w, and changing a wage carries a utility cost of $\frac{\phi_w}{2}(\pi_t^w - 1)^2$, where $\pi_t^w \equiv W_t/W_{t-1}$ is gross wage inflation. An employment subsidy offsets the steady-state wage

markup, so the steady state is the same as in the main text. In a symmetric equilibrium the union's first-order condition is the nonlinear wage Phillips curve

$$\phi_w(\pi_t^w - 1)\pi_t^w = \varepsilon_w L_t C_t^{-\sigma}\Big(\chi L_t^{\eta} C_t^{\sigma} - \frac{W_t}{P_t}\Big) + \beta\frac{\theta_{t+1}}{\theta_t}\,\phi_w(\pi_{t+1}^w - 1)\pi_{t+1}^w, \tag{1}$$

with $W_{-1} = \bar{W}$ and $\pi_t^w = 1$ after the horizon. Wage inflation is positive when the marginal rate of substitution exceeds the real wage, and it depends on expected future wage inflation. Equation (1) replaces $W_t = \bar{W}$. The household's Euler equation, producers, the final-good firm and market clearing are unchanged. Period welfare subtracts the adjustment cost. As $\phi_w \to \infty$ the model becomes the fixed-wage economy; as $\phi_w \to 0$ it becomes the flexible-wage economy.

Calibration. We set $\varepsilon_w = 10$ and choose ϕ_w so that the linearised slope of (1) matches a Calvo wage contract with average duration d months:

$$\frac{\varepsilon_w\,\omega L C^{-\sigma}}{\phi_w} = \frac{(1-\theta_w)(1-\beta\theta_w)}{\theta_w(1+\varepsilon_w\eta)}, \qquad \theta_w = 1 - \frac{1}{d}.$$

We use $d = 12$ months as the baseline, close to the average wage duration in U.S. microdata, and $d = 3$ months as a much more flexible alternative. The slopes are 3.7×10^{-4} and 8.0×10^{-3} per month.

2 Results

Table 1 reports the gain from optimal policy in every experiment, and Table 2 the optimal rate on impact and the share of the peg's loss that policy removes. Figures 1–3 compare the paths.

Table 1: Gain from optimal policy: fixed wage vs. union wages

	time to build			no time to build		
	fixed	union 12m	union 3m	fixed	union 12m	union 3m
3% productivity fall						
Unexpected	0.123	0.123	0.128	0.011	0.012	0.014
News 1 month ahead	0.371	0.371	0.371	0.008	0.008	0.012
News 13 months ahead	0.384	0.385	0.375	0.008	0.009	0.014
News 13 months ahead, policy from $t = 0$	0.106	0.108	0.109	0.001	0.002	0.004
1% demand fall ($\times 10^{-3}$)						
Unexpected	2.18	2.19	2.43	3.65	3.70	4.64
News 1 month ahead	2.68	2.73	3.46	3.65	3.76	5.37
News 13 months ahead	2.36	2.81	3.91	3.65	4.17	6.22
News 13 months ahead, policy from $t = 0$	1.09	1.42	1.77	0.33	0.70	1.39

Gain = optimal − peg, % of one month's steady-state consumption, valued at $t = -13$. "Union 12m" and "union 3m": Rotemberg unions calibrated to 12- and 3-month Calvo contracts.

Table 2: Optimal rate and share of the loss removed

		time to build			no time to build		
		fixed	union 12m	union 3m	fixed	union 12m	union 3m
Productivity	i_0, unexp.	0.0	0.0	0.0	24.9	24.9	24.9
	share, unexp.	31%	31%	32%	100%	100%	100%
	i_{-1}/i_0, 13m	25.4/0.0	25.1/0.0	23.3/0.0	0.0/15.0	0.0/15.0	0.0/14.1
	share, 13m	54%	54%	52%	67%	71%	77%
Demand	i_0, unexp.	2.5	2.5	2.5	0.0	0.0	0.0
	share, unexp.	100%	100%	100%	77%	77%	80%
	i_{-1}/i_0, 13m	4.9/2.6	4.9/2.6	4.9/2.6	12.6/0.0	12.5/0.0	12.1/0.0
	share, 13m	100%	100%	100%	77%	79%	84%

Rates annualised (steady state 4%). "Unexp.": unexpected shock. "13m": shock announced 13 months ahead, policy from the news date. "Share": share of the peg's loss relative to the flexible wage removed by optimal policy.

Twelve-month contracts. The monthly wage barely moves: by less than 0.01% after a 3% productivity fall. With time to build, every productivity gain and optimal rate is within 2% of the fixed-wage value. Without time to build, the gains are two orders of magnitude smaller, and the union moves them more in relative terms. For example, with policy acting only from the shock date the gain is 0.002 against 0.001; with 13 months of warning it is 0.009 against 0.008. For demand shocks with news, the gains are somewhat larger than with the fixed wage (for example 2.8 against 2.4×10^{-3} at 13 months, with time to build). Over a long warning period, unions move wages in anticipation even with sticky contracts, which makes the peg worse. The optimal rate path is the same: evaluated in the fixed-wage model, the union optimum gives the same welfare as the fixed-wage optimum.

Three-month contracts. The comparison with time to build is unchanged where it matters:

- *Sign of the response.* After an unexpected productivity fall, the optimal rate is still cut to zero with time to build and raised to 25% annualised without it.
- *Zero lower bound.* It still binds after the productivity shock with time to build, and for two months after the demand shock without it.
- *Size of the gains.* With time to build the productivity gains hardly change (0.128 unexpected and 0.375 with 13 months of warning). Without time to build they rise, from 0.011 to 0.014. The ratio of gains falls from about 11 to 9 for unexpected shocks, and from 50 to 27 with 13 months of warning. Flexible wages do some of the stabilising work that policy does in the fixed-wage economy, and they help more where policy had less to do.
- *Demand shocks.* The gains rise with wage flexibility in both economies, and the ranking is unchanged: policy matters more for demand shocks without time to build.

Warning and commitment. For productivity shocks, warning still has no value without time to build at either contract length (0.014 with no warning, 0.012 with one month and 0.014 with thirteen, for 3-month contracts). For demand shocks it does once wages are forward-looking: without time to build, the gain rises from 4.6 to 6.2×10^{-3} with 13 months of warning under 3-month contracts. The main text's statement that warning matters only through firms' production plans therefore holds for supply shocks in general, and for demand shocks with a rigid wage.

The commitment exercise, which locks production as if the shock were unexpected, gives:

- productivity, 12-month contracts: 0.1227 (one month of warning) and 0.1229 (13 months), against 0.1228 with no warning;
- demand: 2.21 and 2.31×10^{-3} against 2.19×10^{-3}.

With union wages, expectations alone are worth something, through wages, but only a few per cent of the gain. Firms' re-planning is still almost all of it.

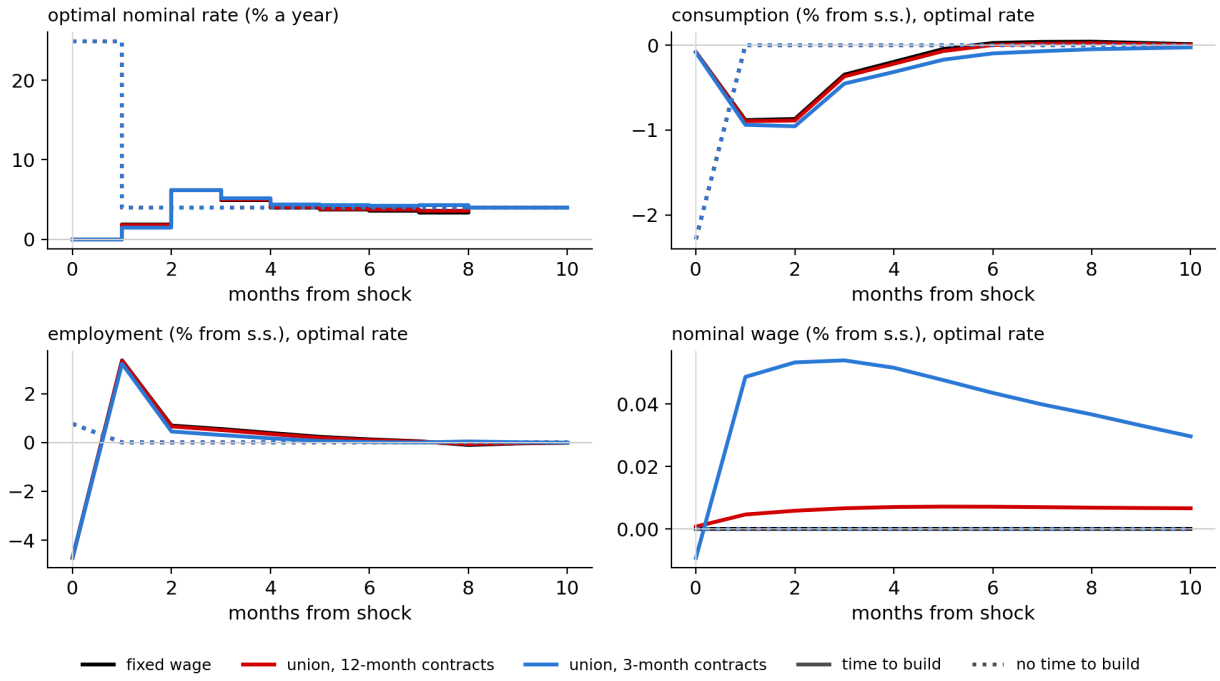


Figure 1: Unexpected 3% productivity fall under the optimal rate: fixed wage (black), unions with 12-month (red) and 3-month (blue) contracts. Solid: time to build; dotted: no time to build.

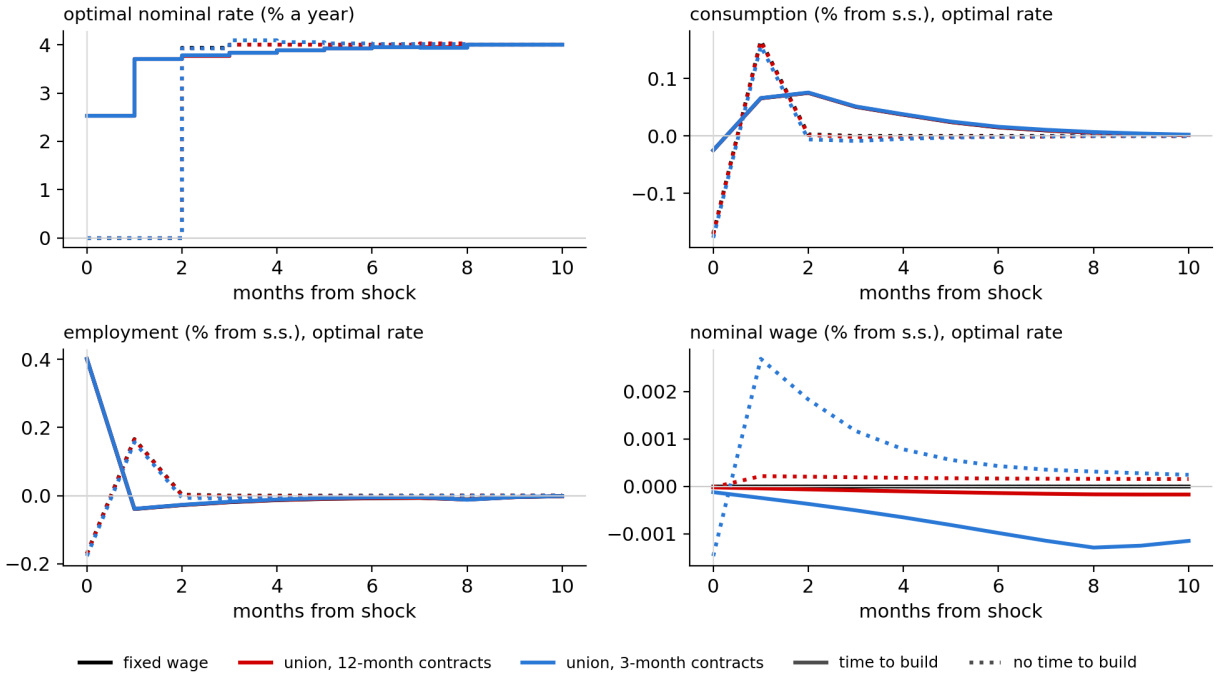


Figure 2: Unexpected 1% demand fall under the optimal rate. Same lines as Figure 1.

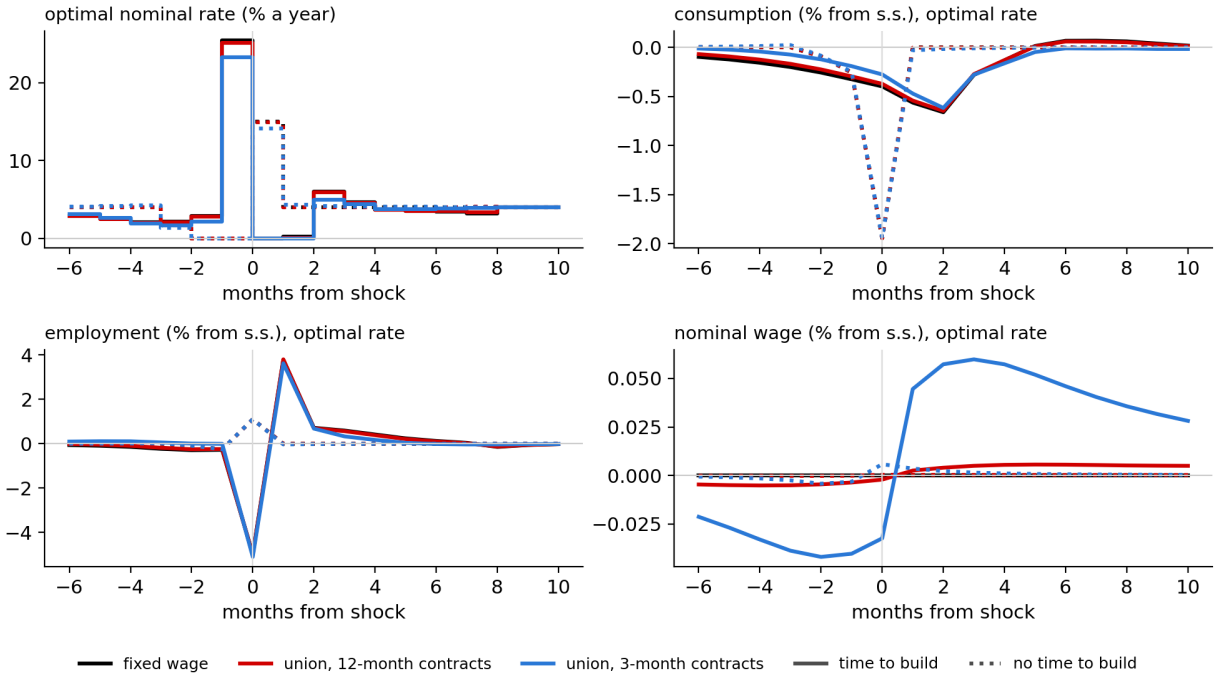


Figure 3: 3% productivity fall announced 13 months ahead, policy from the news date. Same lines as Figure 1.